AG32 LAB MEZZANINE — DIGITAL FABRIC, ADC/DAC, CURRENT MONITOR AND EXPANSION

CONTROL & LAB TOOLS

Ports A0–A5 and N_A0–A5 reach separate AG pins for optional FPGA inline processing. Keep ESP buffers open in inline mode; never oppose drivers.

U14A–U14G are seven sections of ONE physical AG32 chip.

OBS_* pins must remain inputs; fabric reachability, timing and protocol bridges still require HDL/toolchain validation. USB is not sampled by fabric.

EXPAND | ESP32 GPIO

J5: ESP32 + AG32 expansion

ESP GPIO + Qwiic + 4 AG32 signals. 3.3 V only.

GPIO35–37 reserved for PSRAM.

GPIO39–42 share JTAG.

Connects extra modules through a flat cable at the board edge.

PORT SUPERVISOR

AG32 section A | Experiment signals
Reads the ports and handles digital signals.

Observes the ports and manages digital signals for the experiment.

PORT MODE CONTROL

AG32 section B | Port settings
Chooses signal, power or ground for each port.

Chooses how each experiment port is connected.

SUPERVISOR START-UP

20 MHz | Controller clock
Keeps the controller running at a steady pace.

AG32 section F | Power and start-up
Starts the controller and provides programming access.

Powers, starts and provides access to the board controller.

COMMUNICATION HUB

AG32 section C | Board communication
Exchanges data with the computer and other boards.

Links the controller to USB, the network and the optional ESP32.

CURRENT MEASUREMENT

AG32 section D | Current scan control
Chooses which port is measured.

4051 | First measurement selector
Connects one side of the selected current sensor path.

4051 | Second measurement selector
Connects the other side of the same sensor path.

INA226 | Current sensor
Measures the current of the selected port.

Selects one port at a time and measures the current flowing through it.

LOCAL POWER SUPPORT

Keeps the controller supply steady while it works.

SCAN | SMALL TEST SIGNALS

Internal DAC –> R77 –> selectable port. Read both sides of R77 with internal ADC.

Scan: MOSFETs OFF, digital drivers Hi–Z, ESP switches OPEN, R77 bypass OFF.

A return path is required. Occupancy detection is assisted, not guaranteed.

Applies a gentle test signal and watches how each connection responds.

ESP32 PROGRAMS THE AG32

Loads the board controller without an external programmer.

GPIO47 TX / GPIO48 RX | GPIO15 BOOT0 | GPIO16 RESET (open–drain)

AG UART0: TX pin68 / RX pin69. BOOT1 stays LOW via R84.

Recovery: keep ESP32 pins Hi–Z; use UART pads or the existing JTAG pads.

AG Base: U14 fitted, Lab components omitted, R56–R61 = 0R.

Lab: U14 + measurement components fitted. USB remains direct to ESP32.

MEZZANINE INTERFACE — 2 x 60 CONTACTS / MATCH NUMBERS 1:1

Receptacles on carrier, plugs on module. Power off before insertion/removal. Connector footprint and mechanical stack pending layout.

J5 remains the 24–pin flat; J12 exposes remaining ESP contacts. GPIO35/36/37 are isolated by carrier DNP bridges for octal–PSRAM compatibility.

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